VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH MERN

Tech Stack

Online Complaint Registration System — MERN Stack Complaint Management Platform

Phase: Brainstorming & Ideation Phase

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Technology Stack

1. Introduction

This document provides a formal specification of the technology stack selected for the Online Complaint Registration System. Each technology selection is accompanied by its architectural role, implementation purpose, justification, and deployment relevance.

The application is implemented using the MERN stack and follows a modern three-tier architecture consisting of: Presentation Layer, Application Layer, and Data Layer. Additional supporting technologies such as Socket.IO, JWT Authentication, Material UI, MongoDB Atlas, Render, and Vercel are integrated to provide real-time communication, security, scalability, and cloud deployment capabilities.

2. Integration Architecture Diagram

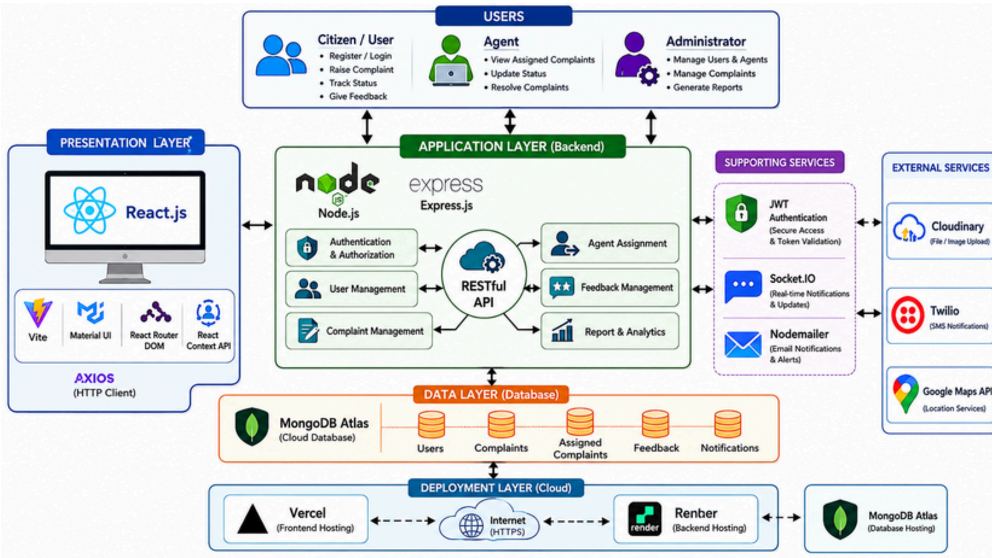

FIGURE 5.1 – TECHNOLOGY STACK INTEGRATION ARCHITECTURE DIAGRAM

3. Detailed Stack Breakdown

Layer	Technology	Version	Purpose & Justification
Frontend Framework	React.js	18.x	Component-based SPA architecture enabling reusable UI development.
Build Tool	Vite	Latest	Fast build system with optimized production performance.
UI Framework	Material UI	Latest	Responsive, modern, accessible UI components.
Routing	React Router DOM	v6	Enables protected navigation for different roles.
State Management	React Context API	Native	Global authentication and app-state management.
HTTP Client	Axios	Latest	API communication with request/response interceptors.

Layer	Technology	Version	Purpose & Justification
Backend Runtime	Node.js	22.x	Asynchronous, scalable event-driven backend service.
Backend Framework	Express.js	5.x	Lightweight REST API framework for middleware architecture.
Authentication	JWT	RFC 7519	Stateless authorization mechanism.
Password Security	BcryptJS	Latest	Industry-standard secure password hashing.
Real-Time	Socket.IO	4.x	Bidirectional messaging and notification channels.
Database	MongoDB Atlas	Latest	Scalable, managed cloud-based NoSQL database.
ODM	Mongoose	8.x	Schema-based MongoDB modeling and validation.
Deployment (FE)	Vercel	Cloud	Edge optimized frontend hosting and CI/CD.
Deployment (BE)	Render	Cloud	Managed backend hosting with auto-deploy.

4. Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web application interface for all stakeholders	React.js, Material UI
2	Application Logic 1	Authentication & Authorization	Node.js, Express.js, JWT
3	Application Logic 2	Complaint Lifecycle Management	Express.js Controllers
4	Application Logic 3	Real-time messaging & chat	Socket.IO
5	Database	Complaint, User, and Message storage	MongoDB Atlas
6	Security Layer	Access control & secure communication	JWT, BcryptJS, RBAC
7	Infrastructure	Cloud hosting platform	Vercel + Render

5. Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	Production-grade open-source libraries	MERN Stack
2	Scalable Architecture	Modular architecture supporting future enhancements	Micro-service approach

S.No	Characteristic	Description	Technology
3	Availability	Cloud-based deployment with high uptime	Vercel, Render, Atlas
4	Performance	Efficient API calls and real-time event updates	Axios, Socket.IO

6. Architecture Decision Records (ADRs)

ADR-01: MongoDB Atlas over Relational Database

Status	✅ Accepted
Context	Dynamic complaint schemas and nested chat/notification data.
Decision	Adopt MongoDB Atlas NoSQL.

ADR-02: JWT Authentication

Status	✅ Accepted
Decision	JWT authentication.
Justification	Stateless, supports cloud deployment scaling.

7. Conclusion

The Online Complaint Registration System utilizes a robust, modern MERN stack. The combination of secure JWT authentication, real-time Socket.IO capabilities, and reliable MongoDB cloud storage provides a foundation for high-quality, maintainable complaint management.